

1. TEST CONFIGURATION

Scenario
synthetic_fs

Target
TGT-01

Camera
640 x 480

FOV
4.0 deg x 3.0

Update Rate
30.0 Hz

Noise: None 10.00 | Camera jitter: OFF 5.00 px/frame | Platform motion: OFF 10.00 px | Turbulence: OFF 10.00 px

2. PERFORMANCE

Processing FPS
301.8
>= 20 FPS

Average Tracking Error
2.68 px

Maximum Tracking Error
6.08 px
≤ 10 px

Acquisition Time
0.03 s
≤ 2 s

Re-acquisition
- s
≤ 1 s

Lock Retention
100.00 %

RMSE
2.97 px

Target Loss
0.00 %

Avg Processing
1.75 ms

Re-acquisitions
0

3. PS REQUIREMENT CHECK

Parameter	Measured	Requirement	Status
Update Rate	30.0 Hz	≥ 20 Hz	PASS
Processing Speed	301.8 FPS	≥ 20 FPS	PASS
Acquisition Time	0.03 s	≤ 2 s	PASS
Tracking Error (Max)	6.08 px	≤ 10 px	PASS
Target Loss	0.00 %	< 5 %	PASS
Re-acquisition	- s	≤ 1 s	-

4. VISUAL SNAPSHOT (T = 1.49 s)

RADAR

Target / Boresight / Prediction

VIRTUAL CAMERA

Target TGT-01 • Error - px

3D DIGITAL TWIN

Camera terminal -> designated beacon

State: BENCHMARK | Pan: -1.17 deg | Tilt: -1.17 deg | Az: -

5. SYSTEM SUMMARY

COARSE ALIGNMENT CONTINUES - HANDOFF NOT READY

Designated target TGT-01 at 1.49 s. Detection, AI verification, Kalman prediction and pan/tilt control are reported from the recorded telemetry snapshot.